

Development Guide (OC)

1. Engineering configuration

- Drag the BluetoothLibrary.framework into the project
- Import header file #import <BluetoothLibrary/BluetoothLibrary.h>

2. Watch scan and connection

2.1 Initialization

CEProduct SDK entry, initialized by subclass (CEProductK6)

```
CEProduct *product = [CEProductK6 sharedInstance];
```

2.2 Scan the watch

Before scanning the watch, check whether the Bluetooth is turned on and whether the permission is set. Receive scanned devices via ScanPeripheralsNoticeKey

2.2.1 Start Scan

```
- (void)startScan;
```

2.2.2 Stop Scanning

```
- (void)stopScan;
```

2.3 Establish a connection

3.1 Establish a connection

```
- (void)connect(CBPeripheral *)peripheral;
```

3.2 Save connected device uuid can be used for automatic connection function

```
- (void)saveConnectedUUID;
```

3.3 Automatic connection

```
- (void)startAutoConnect;
```

3.4 Unbind, and will clear the device information of automatic reconnection

```
- (void)releaseBind;
```

2.4 Connection status detection

2.4.1 Status enumeration

```
searching for ProductStatus_none, ProductStatus_power off, // Bluetooth disconnected
ProductStatus_searching, //Searching
ProductStatus_connecting, //Connecting...
ProductStatus_disconnected, //The connection is disconnected
ProductStatus_connected, //The connection is successful, start pairing
ProductStatus_completed //The pairing is completed, normal communication is possible
```

2.4.2 Detecting the current state

- The attribute
- status notifies ProductStatusChangeNoticeKey to listen for state changes

3. Watch interaction

Instructions are sent using a singleton [CEProductK6 sharedInstance]

```
- (void)sendCmdToDevice:(CE_Cmd *)cmd complete:(void (^)(NSError* error))handler;
```

Receive data

All data is received through this key
CEProductK6ReceiveDataNoticeKey

3.1 App exit

When killing the APP, send the command CE_AppExitCmd to the device without parameters

3.2 OTA upgrade

3.2.1 Get device information

Send command CE_RequestDevInfoCmd, data type: DATA_TYPE_DEVINFO

3.2.2 Get upgrade data

Send command CE_RequestOTAStatusCmd, data type: DATA_TYPE_OTA_STATUS

3.2.2 Upgrading firmware or font library

CE_SendOtaDataCmd

```
- (id)initWithData:(NSData *)otaData otaStatusInfo:
    (NSDictionary*)info
    deviceInfo:(NSDictionary*)deviceInfo
    ompltProgress:(void (^)(CGFloat rate))handler;
```

otaData: Download the local bin file from the server

otaStatusInfo: data received by CE_RequestOTAStatusCmd command

deviceInfo: data received by CE_RequestDevInfoCmd command

compltProgress: upgrade progress value

3.3 Smart camera

CE_SendPhotoCmd

```
- (instancetype)initWithOnoff:(NSInteger)onoff;
```

onoff 0: close take photo; 1: open take photo function

3.4 Sensor switch

It is used to set the device sensor switch. When the APP enters the front and background, it must actively send the data switch status to the device side. The device side decides whether it is possible to upload data to the APP according to the switch status. The device side can upload data only when the switch is turned on. The sensor data is sent to the APP, which can save power

CE_SensorCmd

```
- (instancetype)initWith:(NSInteger)onoff;
```

onoff

When app enter foreground send onoff = 1.

When app enter background or killed send onoff = 0.

3.5 Alarm clock

CE_AlarmItem

Parameter Description:

repeatDay: Alarm repeat day on every day
isRepeatWeek: Repeat alarm every week.
hour: The alarm hour
min: The alarm minute
status: alarm swith status. 0: off, 1: on

CE_SyncAlarmK6Cmd

```
-(nonnull instancetype)initWithItems:(nonnull NSArray<CE_AlarmItem *>*)alarmItems;
```

3.6 Pressure detection

CE_SyncBloodPressureCmd

status: 0 means close detection, 1 means open detection

3.7 Sync custom watch faces

CE_SyncCustomFaceCmd

- **isSupportAccelerate:** If acceleration is supported, the data will be sent at a very high rate. The default value is NO. Use command CE_SyncWatchFaceStartCmd to check whether the device support accelerate.
- **image** Background image. The image size depends on the device. Before transferring the picture data, it needs to be processed into an appropriate size.
- **width** The width of the background image required by the device. Different equipment requires different parameters value.
- **height** The height of the background image required by the device.
- **imageType**
 - 0: Don't change the background image. The image should be null.
 - 1: Use the default background image. The image should be null.
 - 2: Use the image you send. If the image data is null, it will not change the background image.
- **textColor** The custom face text color.
- **timePosition** 0: Time label in the upper part, 1: Time label in the lower part.
- **timeAboveType** Elements shown above time label. 0: Close, 1: Date, 2: Sleep time, 3: heart rate value, 4: step.
- **timeBelowType** Elements shown below time label. The types same as the timeAboveType.
- **imageColorType** Image's color type. 0: RGB565, 1: RGB888

```
-(instancetype)initWithImage:(UIImage *)image width:(CGFloat)width  
height:(CGFloat)height  
  
imageType:(NSInteger)imageType textColor:  
(UIColor *)textColor timePosition:  
(NSInteger)timePosition timeAboveType:  
(NSInteger)timeAboveType timeBelowType:  
(NSInteger)timeBelowType;  
  
-(instancetype)initWithImage:(UIImage *)image width:(CGFloat)width  
height:(CGFloat)height  
  
imageType:(NSInteger)imageType  
textColor:(UIColor *)textColor  
timePosition:(NSInteger)timePosition timeAboveType:  
(NSInteger)timeAboveType timeBelowType:  
(NSInteger)timeBelowType imageColorType:  
(NSInteger)imageColorType;
```

3.8 Do Not Disturb Mode

CE_DisturbItem

- start_time_hour start hour time. 0 to 24
- start_time_min start minute time. 0 to 59

- **end hour time** end hour time
- **end_time_min** end minute time

CE_SyncDisturbCmd

- **switch_flag** Switch status. 0: close , 1: open.
- **disturbItems** disturb setting items. Only support one item in many device.

```
- (instancetype _Nonnull) initWithOnoff:(NSInteger) onoff
                           items:(nonnull NSArray<CE_DisturbItem *>) items;
```

3.9 Target value

CE_SyncGoalCmd

- **step** Step value
- **distance** distance (meter)
- **calories** calories (cal)
- **sleep** sleepTime (minute)
- **sportTime** sportTime (minute)

3.10 Oxygen detection

CE_SyncHeartO2Cmd

- **status** 0: off, 1: on

3.11 Synchronizing data

After the device is connected, send the command CE_SyncHybridCmd to synchronize the basic data,

the basic data includes DATA_TYPE_USERINFO , DATA_TYPE_LANGUAGE_SETTING , DATA_TYPE_TIME , DATA_TYPE_WEATHER , DATA_TYPE_ALARM , DATA_TYPE_SET_TARGET The specific implementation see related instructions

```
CE_SyncUserInfoCmd *userInfoCmd = [[CE_SyncUserInfoCmd alloc] init];
CE_SyncGoalCmd *goalCmd = [[CE_SyncGoalCmd alloc] init];
YD_SyncLanguageCmd *languageCmd = [[YD_SyncLanguageCmd alloc] init];
CE_SyncTimeCmd *timeCmd = [[CE_SyncTimeCmd alloc] init];
CE_SensorCmd *sensorCmd = [[CE_SensorCmd alloc] init];
CE_SyncPairOKCmd *pairCmd = [[CE_SyncPairOKCmd alloc] init];

NSArray *cmds = @[userInfoCmd, goalCmd, languageCmd, timeCmd, sensorCmd, pairCmd];
CE_SyncHybridCmd *cmd = [[CE_SyncHybridCmd alloc] initWithItems:cmds];
[[CEProductK6 sharedInstance] sendCmdToDevice:cmd complete:nil];
```

3.12 Sedentary reminder

CE_SyncLongSitRemindCmd

- **onoff** Switch status. 0: close, 1: open
- **repeatDay** WeekRepeatDay
- **startHour** start hour. 0 to 24.
- **startMinute** start minute. 0 to 59.
- **endHour** end hour
- **endMinute** end minute

```
- (instancetype) initWithOnoff:(NSInteger) onoff weekRepeatDay:
                    (WeekRepeatDay) repeatDay startHour:(NSInteger) startHour
                    startMinute:(NSInteger) startMinute endHour:
                    (NSInteger) endHour endMinute:(NSInteger) endMinute;
```

3.13 Menstrual cycle reminder

CE_SyncMenstrualCycleCmd

- **startTime** Menstrual cycle start time
- **duration** Menstrual cycle duration
- **period** Menstrual cycle period

- **onOff** Reminder switch status. 0: off, 1: on.
- **reminderMintue** Reminder mintue. Values between 0 and 60.
- **reminderHour** Reminder hour. Values between 0 and 24.
- **menstrualNoticeAdvance** One day's advance notice menstrual. 0: off, 1: on.
- **ovulationNoticeAdvance** One day's advance notice ovulation. 0: off, 1: on.
- **ovulationPeakNoticeAdvance** One day's advance notice ovulation peak. 0: off, 1: on.
- **ovulationEndNoticeAdvance** One day's advance notice ovulation end. 0: off, 1: on.

MenstrualCycleType

- **MenstrualCycleTypeNone**
- **MenstrualCycleTypeNonPregnancy** Menstrual cycle at non-pregnancy period
- **MenstrualCycleTypePregnancy** Menstrual cycle at pregnancy period #### 3.14 Pairing

CE_SyncPairOKCmd

Used to confirm to the device side whether the pairing is complete. When the APP is successfully connected and bound with the device (such as scanning the QR code or entering the pairing number), the APP should actively notify the device side, and the device side will perform corresponding operations (such as exiting the QR code interface) according to the pairing status information. , or save related pairing information, etc.), this command uses the data type DATA_TYPE_PAIR_FINISH

Parameters firstPairStatus: 0: normal, 1: first times pair

3.15 Synchronized system watch face

CE_SyncSystemFaceDataCmd

- **isSupportAccelerate** If acceleration is supported, the data will be sent at a very high rate. The default value is NO. Use command CE_SyncWatchFaceStartCmd to check whether the device support accelerate.

```
CE_SyncSystemFaceDataCmd *watchFaceDataCmd = [[CE_SyncSystemFaceDataCmd alloc] init]; watchFaceDataCmd.data =
fileData; watchFaceDataCmd.isSupportAccelerate = NO; watchFaceDataCmd.syncProgress = ^(CGFloat progress)
{ dispatch_async(dispatch_get_main_queue(), ^{

    // do something

});
};
[CEProductK6.sharedInstance sendCmdToDevice:watchFaceDataCmd complete:^(NSError *error) {
    // do something
}];
```

CE_SyncWatchFaceStartCmd

Determine whether the device supports high-speed data transmission mode. The current data types supported by the high-speed data mode are

CE_SyncSystemFaceDataCmd , CE_SyncCustomFaceCmd .

3.16 Set time

CE_SyncTimeCmd

- **absTime** timestamp since 1970
- **offsetTime** time zone offset
- **format** 0:12-hour, 1:24-hour
- **mdFormat** 0: month-day, 1: day-month

```
- (instancetype)initWithAbsTime:(double)absTime
                        offset:(NSInteger)offsetTime format:
                        (NSInteger)format mdFormat:
                        (NSInteger)mdFormat;
```

3.17 Synchronize user information

CE_SyncUserInfoCmd

- **sex** 1: female 0:
- **male IrHand**: 1: right hand 0: left hand

3.18 App message notification settings

CE_SyncWatchMenuCmd

- **allSwitch** Master switch. If master switch is disabled, all of the message alert will be closed.

```
CE_SyncWatchMenuCmd *watchCmd = [[CE_SyncWatchMenuCmd alloc] init];
watchCmd.allSwitch = YES;
watchCmd.qq = YES ;
watchCmd.email = YES ;
watchCmd.wechat = YES ;
watchCmd.facebook = YES;
watchCmd.twitter = YES;
watchCmd.whatsapp = YES;
watchCmd.instagram = YES ;
watchCmd.skype = YES;
watchCmd.linkedin = YES;
watchCmd.line = YES;
watchCmd.kakaoTalk = YES;
watchCmd.telegram = YES;
watchCmd.other = YES ;
[[CEProductK6 sharedInstance] sendCmdToDevice:watchCmd complete:^(NSError *error) {

}];
```

3.19 Synchronized Weather

CE_WeatherItem

- **weather**

New weather type code

code	mean
10	sunny
11	partly cloudy
12	cloudy day
13	shower array
14	Thundershower
15	Light rain small
16	Moderate rain
17	heavy rain
18	SleetySnow
19	Light snow
20	Moderate snow
weather code	The heavy snow
weather code	haze haze
weather code	The wind

deprecated below:

Old version of weather code

code	mean
0	sunny
1	cloudy
2	partly_cloudy
3	rain
4	snow

- **low_temperature**

The lowest temperature value of the day. Unit of value is Celsius.

- **high_temperature**

highest temperature value of the day. Unit of value is Celsius.

- **PM**

air quality value. Could ignore this value.

CE_SyncWeatherCmd

- **time**

Current time's timestamps

- **todyWeather**

today's weather

- **tomorrowWeather**

tomorrow's weather

- **dayAfterTomorrowWeather**

the day after tomorrow's wether

3.20 Unit setting

CE_UnitSettingCmd

- **weight**

weight unit. 0: kilogram, 1: pound, 2: stone

- **distance**

distance unit. 0: meters, 1: miles

- **weather**

weather unit. 0: Celsius, 1: Fahrenheit

3.21 Selecting the watch face

YD_SyncFaceIndexCmd

/// initialize sync face index command /// A device generally has several dials, depending on the device.
/// The watch face can be switched by sending the face index. /// @param index Index of the watch face you chose.

3.22 Synchronized motion

YD_SyncSportCmd Application workout status set from device

- **type**

Workout type. Define your own.

- **status**

Workout status. See enum SPORT_STATUS_SYNC.

- **time**

Workout time with unit seconds.

- **distance**

Workout distance.

SPORTSTATUSTYPE

SPORT_STATUS_STOP = 0, // Stop workout
SPORT_STATUS_START, SPORT_STATUS_PAUSE, // Start or PAUSE
SPORT_STATUS_CONTINUE, // Resume workout
SPORT_STATUS_STOP_FORCE, // Force stop workout
SPORT_STATUS_SYNC, // Needed send app workout data to device. You can't send this type to device.

3.23 Mobile Contacts

ContactCmdType

- name

contact name

- phone

mobile phone number

- cmdType

See ContactCmdType for details

- removeIndex

drop index

- syncIndex

Synchronized index

ContactCmdType

ContactCmdTypeAdd, //add
ContactCmdTypeDelete, //delete
ContactCmdTypeClear, //clear
ContactCmdTypeSync //synchronous

Sync contacts, data type DATA_TYPE_CONTACT_SYNC

```
ContactCmd *cmd = [[ContactCmd alloc] init]; cmd.cmdType =  
ContactCmdTypeSync; cmd.syncIndex = index; [[CEProductK6  
shareInstance] sendCmdToDevice:cmd complete:nil];
```

Note that contacts are synchronized one by one

3.24 Real-time heart rate detection

CE_SyncHeartRateCmd

- status

// 0: close, 1: open

3.25 Automatic Heart Rate Setting

YD_SyncAutoHeartCmd

auto heart rate detection setting command

- onoff

/// Automatic heart rate detection. Once every 10 minutes. /// 0: close, 1: open.

- hr24hOnoff

/// Heart rate measurement for 24 hours
/// Valid when DATA_TYPE_FUNCTION_CONTROL hasHR24h is true

3.25 Incoming call settings

YD_SyncCallAlarmCmd

- onoff

onoff Switch status. 0: close, 1: open.

3.26 Drinking water reminder

YD_SyncDrinkAlarmCmd

- onoff

Switch status. 0: off, 1: on

- startHour

startHour. 0-24

- startMintue

startMintue. 0-59

- endHour

endHour

- endMinute

endMinute

- intervalLevel

drink water reminder interval. Four level support: 0: 30 minutes, 1: 1 hour, 2: 2 hours, 3: 3 hours

3.27 Find Devices

YD_SyncFindDevCmd

- onoff

onoff Switch status. 0: close, 1: open.

3.28 Goal Achievement

YD_SyncFinishGoalCmd

- onoff

onoff Switch status. 0: close, 1: open.

3.29 Language settings

YD_SyncLanguageCmd

language
0: English(English) 1: Simple-Chinese(Simplified Chinese), 2: Spanish(Western Class) 3: Italian(Italian),
4: Portuguese(Grape), 5: French(French), 6: Japanese(Japanese), 7: Russian(Russian), 8: Korean(Korean),
9: German (German), 10: Traditional-Chinese (Traditional Chinese), 11: Arabic (Arabic) 12: Indonesian (Indonesian),
13: Turkish (Turkish), 14: Ukrainian (Ukrainian), 15: Hebrew (Hebrew), 16 Polish, 17 Hindi, 18 Croatian

3.30 SMS reminder

YD_SyncSMSAlarmCmd

- onoff

onoff Switch status. 0: close, 1: open.

3.31 Raise your wrist to brighten the screen

YD_SyncUpBrightCmd

- onoff

Switch status. 0: off, 1: on.

- startHour

start hour. 0 - 24

- startMinute

start minute. 0 - 59

- endHour

end hour. 0 - 24

- endMinute

end minute. 0 - 59

3.32 Get alarm clock information

CE_RequestAlarmInfoCmd

- DATA_TYPE_ALARM

3.33 Get Mixed Data

CE_RequestAllInfoCmd

- DATA_TYPE_DEVINFO
- DATA_TYPE_BATTERY_INFO
- DATA_TYPE_SETTING
- DATA_TYPE_ALARM

3.34 Get device power data

CE_RequestBatteryCmd

- DATA_TYPE_BATTERY_INFO

3.35 Get device information

CE_RequestDevInfoCmd

- DATA_TYPE_DEVINFO

3.36 Get Do Not Disturb Mode Information

CE_RequestDisturbCmd

- DATA_TYPE_FORGET_DISTURB

3.37 Get target data

CE_RequestGoalCmd

- DATA_TYPE_SET_TARGET

3.38 Get sedentary reminder data

CE_RequestLongSitCmd

- DATA_TYPE_SITTING_REMIND

3.39 Request pairing information

CE_RequestSystemPairStatusCmd

- DATA_TYPE_BLE_PAIR_STATUS

3.40 Requesting User Information

CE_RequestUserInfo

- DATA_TYPE_USERINFO

3.41 Get dial information

YD_RequestWatchFaceSettingCmd

- DATA_TYPE_WATCH_FACE_INFO